

Towards Learning-Based Bathymetry Reconstruction using Forward-Looking Sonar Images

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Abstract—Forward-looking sonar (FLS) is widely used on autonomous underwater vehicles (AUVs) for seafloor mapping and collision avoidance. However, it usually has a wide vertical aperture. As a result, elevation information is unavailable because echoes returned from different elevation angles at the same range and azimuth will merge into a single image pixel. This will make terrain reconstruction challenging. To this end, this paper presents a three-stage convolutional neural network to predict per-pixel elevation angles from a FLS image for 3D cloud generation. The network is trained using co-registered multibeam echosounder (MBES) measurements. The staged decomposition is designed to perform no-return rejection, partial-return classification and elevation regression. It is motivated by the fact that 86.5 % of pixels in an FLS image have no valid returns. The elevation regression stage is trained with a locally weighted Huber loss function normalized by the total-variation and Laplacian smoothness terms along each beam. The method is validated using the simulated data generated in the Stonefish simulator. The pipeline attains an F1 score of 86.3 % on the classification (stage 1&2), an RMSE of 1.03 deg (0.018 rad) in simulation and RMSE of 6.51 deg (0.11 rad) deg on the field dataset. To make the trained model applicable to field data, speckle noise was applied on the simulated sonar images and bilateral filtering was applied with intensity thresholding to enhance the simulated images that are used in model training. Then the trained model is applied to the field data to create a point cloud of the terrain and compared with the results collected using a Norbit MBES mounted on an ASV.

Index Terms—Forward-looking sonar, imaging sonar, bathymetry, convolutional neural networks, underwater perception.

I. INTRODUCTION

Reconstruction of the seafloor terrain is crucial for underwater navigation, seabed mapping and hydrographic surveys. Forward-looking sonar (FLS), which is increasingly used on autonomous underwater vehicles (AUVs), provides high-resolution acoustic images [1] making it suitable for collision avoidance, target detection, and simultaneous localization and mapping. A typical FLS has a wide elevation aperture (10-20 degrees), causing geometric ambiguity. The elevation angle of a return cannot be resolved, and the corresponding pixel in the sonar image may be resulted from multiple returns from different elevation angles at the same range [2]. FLS therefore cannot be used directly for terrain reconstruction.

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In this paper we present a framework that addresses the ambiguity using a multi-stage neural network trained with multibeam echosounder (MBES) data. The contributions are as follows.

- 1) A three-stage convolutional architecture is introduced that will separate no-return rejection, partial-return classification and elevation regression, to deal with the extreme class imbalance of FLS data.
- 2) A regression objective function is proposed that combines a Huber loss normalized by local elevation variation and Laplacian smoothness to preserve features.
- 3) A label-generation procedure is also proposed that re-projects MBES soundings into the FLS image plane to handle the explicit field-of-view and incidence-angles. This proposed procedure encodes no-return and partial-return cases as distinct label classes.

The model is trained on a simulated dataset and then validated on a test set (not used in training). Further, the model has been applied to a field dataset to inference elevation angles for 3D point cloud generation.

II. RELATED WORK

Geometric and multi-view methods: Prior work resolves elevation through multi-view fusion [2], [3], space carving [3], occupancy grid formulations [4], or acoustic structure from motion. These methods recover accurate geometry when the same terrain is imaged from several viewpoints with known relative pose, and are the established route to dense reconstruction from FLS. Their common requirement of multiple overlapping passes increases survey time, makes performance dependent on pose accuracy across the sequence, and needs single-frame inference from an AUV moving through unseen terrain.

Neural scene representations: Neural networks have been applied to AUV navigation in a range of ways [5]. Two recent lines apply implicit neural representations to acoustic data, differing in what they reconstruct and how the forward model is handled. The first targets object-scale geometry, representing the scene implicitly and inverting a differentiable acoustic renderer to reconstruct objects at high fidelity from multi-view FLS imagery [6]; the renderer encodes the sonar imaging physics, so reconstruction quality is tied to how well that model matches the deployed sensor. The second targets bathymetry, modelling the seafloor as a neural heightmap and

estimating the beam patterns jointly with the terrain, which relaxes the requirement for a pre-calibrated sensor model and suits large-area mapping rather than isolated targets [7]. Both are self-supervised and therefore need no labelled elevation, and between them they represent the strongest reconstruction quality currently reported for FLS. Both also share the structural limitation of the family: the result is a scene, not a predictor. Each new site requires its own optimization, and each surface requires enough overlapping observations to be constrained, so neither supports single-frame inference during a first pass.

Single-view learned elevation estimation. Supervised methods enable single-frame prediction [8], [9], and related work regresses an equivalent pseudo front-view depth from one acoustic frame [10], [11]. This line of work shows that a network trained on paired data can resolve the elevation ambiguity from a single frame, removing the multi-view and per-scene requirements of the approaches above. Existing methods are trained and evaluated primarily on object-scale targets rather than seafloor bathymetry, do not model the class imbalance between no-return, partial-return and valid positions that dominates FLS imagery, and are not solely supervised by MBES soundings, which is the gap the present work addresses.

III. PROBLEM FORMULATION

A. FLS Imaging Geometry

Any physical point or surface element that reflects the transmitted acoustic wave back toward the receiver in the sonar frame can be represented in the polar coordinate (r, θ, ϕ) with range (r) , azimuth angle (θ) and elevation angle (ϕ) , and converted into a point in the Cartesian space using the following equation.

$$\mathbf{p} = [r \cos \phi \cos \theta \quad r \cos \phi \sin \theta \quad -r \sin \phi]^\top. \quad (1)$$

From a 2D FLS image, we can obtain the range and azimuth information, but no elevation angle is available. Each pixel in the image represents the echo intensities from the elevation arc $\mathcal{A}(r_i, \theta_j)$ defined as:

$$\mathcal{A}(r_i, \theta_j) = \{ \mathbf{p}(r_i, \theta_j, \phi) : \phi \in [-\phi_{\max}, \phi_{\max}] \}, \quad (2)$$

where $\phi_{\max}$ is 6 deg for the Oculus M750d running at 1.2MHz used in our case. Recovering terrain requires estimating the elevation angles for each pixel.

B. Label Generation from MBES

We assume an MBES is installed with its swath overlapping the elevation field of view (FOV) of the FLS. Each MBES sounding is transformed into the FLS sonar frame, converted into (r, θ, ϕ) , and binned by FLS range index. We designed up to four intersections permitted per range bin. When several MBES soundings fall into the same range bin they are sorted by incidence angles in increasing order under the Lambertian scattering assumption, because echo intensity from a smaller incidence angle is higher.

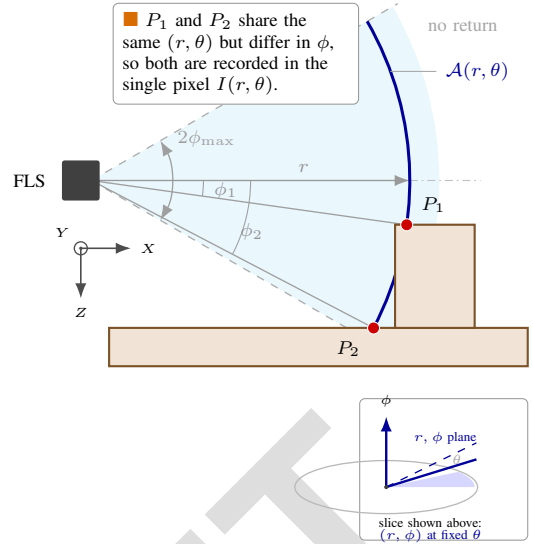


Fig. 1: Elevation ambiguity in FLS imaging. The main diagram shows the elevation plane at fixed azimuth θ : every scatterer on the iso-range arc $\mathcal{A}(r_i, \theta_j)$ projects to one image pixel, so the returns P_1 from the raised structure and P_2 from the seafloor are indistinguishable in the sonar image. Most of the arc lies in open water and produces no return, which is the source of the class imbalance addressed in section IV-A. The inset shows this (r, ϕ) plane's position within the full spherical (r, θ, ϕ) frame: θ sweeps out of the page and is not shown to scale in the main diagram.

Each position in the label array has three conditions: a no-intersection flag (condition 1), 1-3 intersection angles (condition 2), and 4 intersections (condition 3). For condition 1, all four elevation angles are encoded as -20 radians. For condition 2, the places with no return are encoded as -10 radians.

IV. APPROACH AND METHODOLOGY

As shown in the pipeline diagram (Fig. 2), the elevation angle prediction task is decomposed into three sequential stages. Each one is implemented as a one-dimensional CNN whose input is the sequential image intensities at increasing range bins along a single beam (one column in the raw FLS image).

A. Stage 1: No-Return Detection

A convolutional encoder with transposed-convolution up-sampling is designed to identify the occurrence of condition 1, the most frequently observed for a range bin (86.5% in our case). Once these data points are identified, we can remove them for future stages. The training loss of the network is defined in Eq. 3 based on weighted binary cross entropy [12]. We define a weight, w_1 , being a ratio of the number of data points in condition 1 and the sum of the number of data points in condition 2 and 3 where the ratio is clamped between 1 and 10. The weight, w_1 , is recomputed per training batch

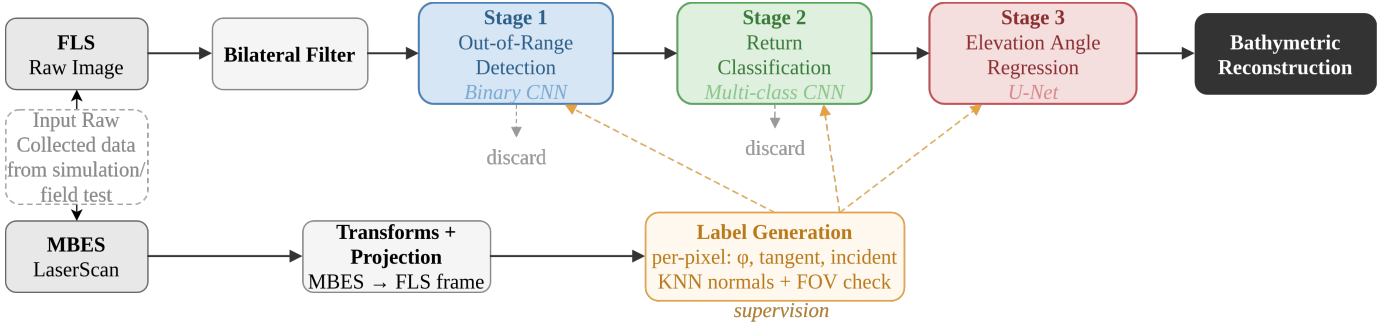


Fig. 2: The three-stage pipeline predicting elevation angles for bathymetric reconstruction, with ground truth derived from MBES soundings reprojected into the FLS image plane.

rather than fixed at this dataset-wide value, so realized per-batch values average around 6.42. We also define t_i as the ground-truth binary label that equals 1 for a range bin under condition 1, and P_i is the output of the network, the predicted probability of condition 1 for the i -th range bin.

$$\mathcal{L}_{s1} = -\frac{1}{N} \sum_{i=1}^N [(1 - t_i)P_i + w_1 t_i(1 - P_i)] \quad (3)$$

B. Stage 2: Valid versus Partial Return Classification

Stage 2, a CNN encoder network, is designed to identify conditions 2 and 3 for a range bin. A mask will be created from the classification result and be used in Stage 3 for predicting the elevation angles. The stage 2 network is trained using a similar loss function presented in Eq. 3. However, t_i becomes the ground-truth binary label that equals 1 for a range bin under condition 2. The weighting w_i is calculated similar to stage 1, which is a ratio between the number of data points in condition 2 and condition 3.

C. Stage 3: Elevation Regression

In stage 3, a U-Net [13] encoder-decoder with a dilated bottleneck [14] and residual blocks is designed. The network will predict elevation angles for the range bins in conditions 2 and 3. Predictions are reshaped to $\hat{\Phi} \in \mathbb{R}^{B \times N_b \times N_r}$ for the smoothness terms, where N_b and N_r are the numbers of beams and range bins. Each point is weighted by the local variation of the ground truth elevation angles over a window, $W(i)$, that includes K number of neighboring bins. For a neighbor bin $j \in W(i)$, the ground-truth elevation angle is denoted as y_j . Its weight is computed using Eq. 4 where γ is a tuning parameter. For the final Loss function, we multiply Huber loss with this weight, thus highly varying regions will result in higher weight, as shown in Eq. 7.

$$w_i = 1 + \gamma \frac{\sum_{j \in W(i)} |y_j - y_i|}{K}, \quad (4)$$

Two additional smoothness terms are added in the loss function shown in Eq. 7. First, we calculate the total variation, $\mathcal{L}_{TV}$, to penalize the discrete first derivative between adjacent valid data points. The equation is shown in Eq.5, where the valid adjacent pairs used by the total-variation terms are along

the range ($\mathcal{M}_r$) and along the beams ($\mathcal{M}_b$). Second, we compute a Laplacian term to penalize the discrete second derivative (curvature) between three data points using Eq.6 along range and beams. These terms are calculated for each individual sonar frame and averaged over the batch.

$$\mathcal{L}_{TV} = \frac{1}{|\mathcal{M}_r|} \sum_{\mathcal{M}_r} |\hat{\Phi}_{k,j+1} - \hat{\Phi}_{k,j}| + \frac{1}{|\mathcal{M}_b|} \sum_{\mathcal{M}_b} |\hat{\Phi}_{k+1,j} - \hat{\Phi}_{k,j}|, \quad (5)$$

$$\mathcal{L}_{lap} = \frac{1}{|\mathcal{M}_r^2|} \sum_{\mathcal{M}_r^2} (\hat{\Phi}_{k,j+1} - 2\hat{\Phi}_{k,j} + \hat{\Phi}_{k,j-1})^2 + \frac{1}{|\mathcal{M}_b^2|} \sum_{\mathcal{M}_b^2} (\hat{\Phi}_{k+1,j} - 2\hat{\Phi}_{k,j} + \hat{\Phi}_{k-1,j})^2, \quad (6)$$

$$\mathcal{L}_{s3} = \frac{\sum_{i \in \mathcal{V}} w_i \mathcal{L}_\delta(\hat{\phi}_i - y_i)}{\sum_{i \in \mathcal{V}} w_i} + \lambda_{tv} \mathcal{L}_{TV} + \lambda_s \mathcal{L}_{lap}, \quad (7)$$

The parameters in the loss function are set as follows in our training. $K = 5$, $\gamma = 1.0$, $\lambda_{tv} = 0.005$ and $\lambda_s = 0.0005$.

During the inference the accepted probabilities for condition 1 in Stage 1 are set to 0.5, and for condition 2 in Stage 2 is 0.65. These probability thresholds were found to yield the highest F1 score in Stage 1 and 2.

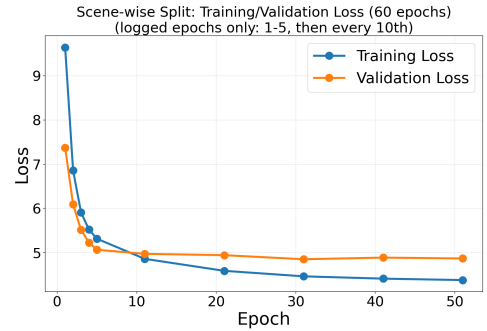


Fig. 3: Training and validation loss over 60 epochs on the data split (combined Stage 1 + Stage 2 + Stage 3 objective).

D. Training and Inference

The three stages are trained using separated AdamW [15] optimizers with OneCycleLR [16] scheduling (pct_start =

0.3, weight decay 10^{-4}). Stage 1, 2 and 3 use a peak learning rate of 10^{-4} , 10^{-4} , and 10^{-5} , respectively. All three are trained for 60 epochs with a batch size of 4. Parameter counts are 20,961 (Stage 1), 19,089 (Stage 2) and 3,884,929 (Stage 3). Training was performed on a NVIDIA GeForce RTX 4070 SUPER (12 GB VRAM). All three networks are training together with a total training time of about 30 minutes, with training and validation loss shown in Fig. 3. We observe fast decline in loss during the first 10 epochs, then validation loss converge while training loss continues to decrease slowly.

V. EXPERIMENTAL SETUP

A. Simulated Dataset

Simulated data were collected in a tank environment using the Stonefish simulator [17]. Different objects, such as boxes, cylinders, ladders, spheres and stair structures, as shown in Fig. 4, are placed in the scene. We collected a total of 14,297 FLS images with aligned MBES soundings.

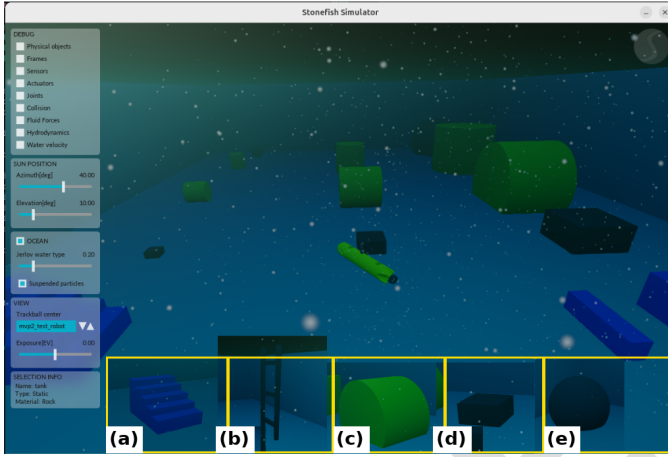


Fig. 4: The simulated environment in Stonefish with the available objects shown in sub-figures a) stairs, b) ladder, c) cylinder, d) box, e) sphere.

We have excluded the FLS images without any objects when the vehicle is transiting between objects. Then, the remaining images are randomly split into three groups: training (70%), validation (20%) and test (10%).

B. Field Dataset

Field data were collected using ALPHA AUV [18] equipped with a Blueprint Oculus M750d FLS which moved around the Whale Rock in Narragansett Bay. The reference bathymetry was obtained from an ASV-based survey using Norbit MBES. It is worth mentioning that the FLS was oriented such that its elevation angle is aligned along vehicle's longitudinal direction to obtain wide vertical swath. This setup is different from the simulation environment.

C. Noise Modeling and Pre-processing

To make the model capable of predicting elevation angle from field data set, we purposely tuned the speckle noise in the

simulation to match the noise in field-collected FLS images. A pre-processing method is created to remove the noise (see Fig. 5). First, a bilateral filter [19] is applied to suppress the speckle while preserving the acoustic edges. Then, we applied intensity thresholding to remove low-amplitude artifacts. In the field data, we have observed returns from water surface, and thruster disturbance. Therefore, we only select a region of interest from each image to be used for inference and terrain reconstruction. Note that this region is subject to change if the sonar or sonar orientation has changed.

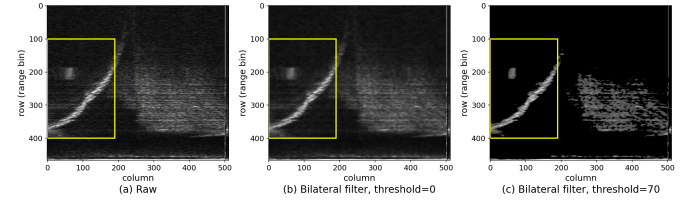


Fig. 5: Selected field FLS frame under different denoising settings: (a) raw sonar image, (b) bilateral filter, and (c) bilateral filter with binary threshold = 70. The yellow box marks the region of interest used for terrain reconstruction on every image frame.

D. Metrics

Reconstruction quality is evaluated by nearest-neighbor distance between the predicted and MBES point clouds in 3-D Euclidean space using a KD-tree [20]. Correspondence is reported as the fraction of predicted points with an MBES point within the acceptable tolerance (τ) between MBES and FLS points in 3D. We additionally report elevation-angle RMSE and MAE in degrees, computed between the network's predicted elevation angle and the reprojected MBES data.

VI. RESULTS

A. Per-Object Analysis in simulation

The reconstruction results on two objects, a cylinder and a ladder, are shown in Fig. 6, comparing the point cloud generated by the network from FLS images, the MBES reference and the per-point nearest-neighbour distance. For the cylinder, we observed low coverage on its surface around the top may be due to lack of data points from MBES. For the ladder, the point cloud generated from FLS images successfully rendered the two vertical poles and the top horizontal bar. However, the lower horizontal bar was missing.

In Table I, we list the correspondence for increasing nearest neighbor thresholds. Correspondence increases sharply between $\tau = 0.03$ m and $\tau = 0.07$ m, indicating that most of the distance errors between the MBES points and the points generated from FLS images lie in that interval. We also found objects with planar surfaces, such as the stairs and ladder, achieve higher correspondence than objects with curved geometry such as the cylinder and sphere.

To investigate the impact of surface complexity on our result, we looked into per frame prediction of elevation angles.

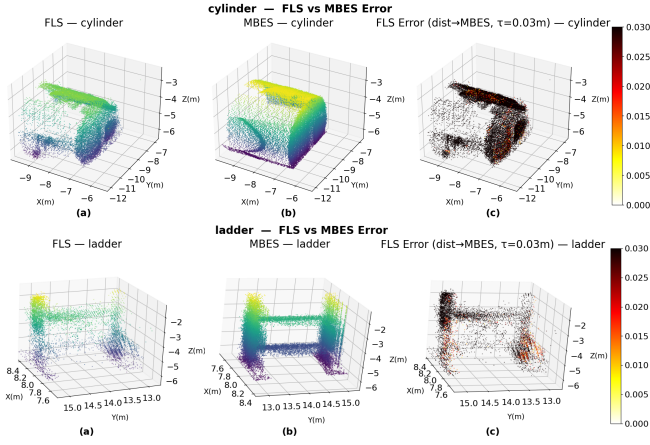


Fig. 6: FLS reconstruction (a) against MBES ground truth (b) for a cylinder (top) and ladder (bottom) at $\tau = 0.03\text{m}$. Distance errors between the two sources are computed by mean nearest neighbors and shown in (c).

The frames are split into Low (less than 0.037), Medium (0.037 to 0.149) and High curvature (higher than 0.149). The regions of curvature values are calculated over the full dataset. Fig. 7 shows an example frame from each category. It can be seen that as curvature increases, the 2D geometry (red) created using the predicted elevation angles from the network diverges from the real values (blue).

TABLE I: Correspondence between the two point cloud sources on different objects at increasing nearest-neighbour thresholds.

Object	Correspondence (%)			
	$\tau=0.01\text{m}$	$\tau=0.03\text{m}$	$\tau=0.07\text{m}$	$\tau=0.10\text{m}$
Box	0.2	5.4	34.6	53.7
Cylinder	0.2	3.1	18.8	31.9
Ladder	1.0	17.1	59.3	83.1
Sphere	0.2	4.5	32.3	52.2
Stairs	7.1	42.5	86.2	95.9

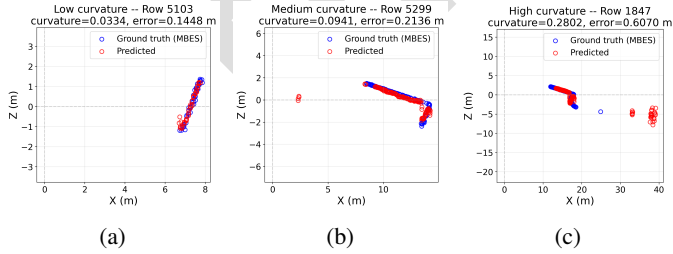


Fig. 7: Example reconstructed 2-D profiles by curvature ter-cile: (a) Low, (b) Medium, (c) High. Each panel shows ground-truth MBES points (blue circles) and predicted points (red circles) in the sensor (X-Z) plane for one representative test frame from that class.

We further present the histogram of the distance errors of the per-frame point clouds between prediction and actual

values in Fig. 8 where a similar trend is observed. A plausible explanation is that a high-curvature profile spans a wider range of incidence angles within a single beam. This leads to a weakened Lambertian ordering assumption used in label generation (Section III-B) and increases the number of range bins receiving multiple intersections.

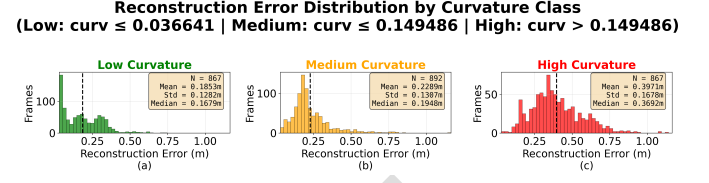


Fig. 8: Distribution of per-frame mean reconstruction error for the test set ($n = 2,626$), split into curvature terciles: (a) Low ($\kappa \leq 0.037$), (b) Medium ($0.037 < \kappa \leq 0.149$) and (c) High ($\kappa > 0.149$). Dashed lines mark the class means. Error grows with curvature in mean, median and spread, and the Low and Medium classes are right-skewed while the High class is broader and closer to symmetric.

B. Application to Field Data

We applied our network to the field data without retraining. Table II reports the overall correspondence between reconstructed points from FLS images and the ground truth MBES data. Only a short section of the bathymetry with steep walls was used for better comparison. For the selected portion, the coverage rises from 16.8% ($\tau = 0.10\text{m}$) to 39.5% ($\tau = 0.20\text{m}$) roughly half the corresponding coverage on simulated data (Table I). Fig. 9 shows the reconstructed cloud and the MBES reference at $\tau = 0.2\text{m}$, where 39.5% of FLS points are matched to MBES data points. For the same $\tau = 0.20\text{m}$ correspondence points, the elevation-angle error between the predicted and reprojected ground-truth angles gives an RMSE of 6.51 deg and an MAE of 5.39 deg.

TABLE II: FLS vs. MBES reconstruction metrics across object geometries.

Object	Correspondence (%)		
	$\tau=0.10\text{m}$	$\tau=0.15\text{m}$	$\tau=0.20\text{m}$
Whale Rock (carved)	16.8	29.7	39.5

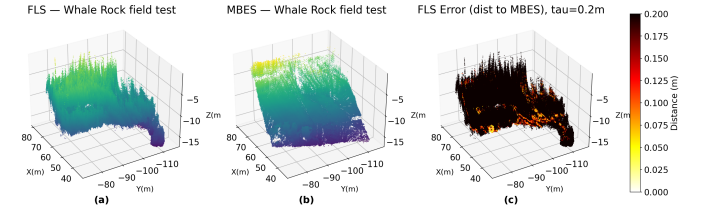


Fig. 9: (a) FLS reconstruction, (b) MBES reference for a region-of-interest carved from the Whale Rock field test at $\tau = 0.20\text{m}$, (c) Distance errors computed by mean nearest neighbours.

C. Runtime

The network has almost 4 million parameters, 15.7 MB at single precision. Table III reports per-frame cost for the complete single-column pipeline, measured on 120 sonar images with pixels of 467×512 , including bilateral filtering with intensity thresholding, elevation inference, and projection to Cartesian coordinates. The test was performed on AMD Ryzen™ 9 7950X3D CPU.

TABLE III: Per-frame runtime, single-column inference, measured on field imagery.

Stage	Time (ms)
Image decode	0.02
Bilateral filter and threshold	2.21
Elevation inference	5.24
Backprojection	0.02
Total (CPU (AMD Ryzen), 4 threads)	7.53
Total (GPU)	2.20

VII. CONCLUSION AND FUTURE WORKS

We presented a three-stage convolutional pipeline that resolves the elevation ambiguity of forward-looking sonar from a single frame, supervised by co-aligned multibeam-returned data. The pipeline combines a locally weighted Huber loss with total-variation and Laplacian smoothness terms, and a label-generation procedure that reprojects MBES soundings into the FLS image plane. The results have shown an approximate 60% correspondence with simulation and 39.5% for a selected field test dataset. Future work will address cross-frame fusion with predicted uncertainty, the implementation on embedded hardware for real-time inference in the field, and collecting more real field datasets to test. Improvement on the results and training on one field dataset and inference on another field dataset will also be a part of future work, along with investigating the elevated error observed on high-curvature geometry.

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